

Cong Cao

PhD Student in Computer Vision

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[Google Scholar](#) | [GitHub](#)

RESEARCH FOCUS

3D generative modeling and geometry learning for digital humans, with applications in garment creation, virtual try-on, and simulation-ready assets.

EDUCATION

Mohamed bin Zayed University of Artificial Intelligence (MBZUAI)

PhD in Computer Vision

Advisor: **Prof. Hao Li**

📅 Aug. 2024 – Present

📍 Abu Dhabi, UAE

Mohamed bin Zayed University of Artificial Intelligence (MBZUAI)

MSc in Computer Vision

Advisor: **Prof. Hao Li**

📅 Aug. 2022 – Jun. 2024

📍 Abu Dhabi, UAE

University of Bristol

MSc in Robotics | Graduated with Distinction

Advisor: **Dr. Paul O'Dowd**

📅 Sep. 2021 – Nov. 2022

📍 Bristol, UK

Xidian University

BEng in Intelligence Science and Technology

GPA: 3.8/4.0

📅 Sep. 2018 – Jul. 2022

📍 Xi'an, China

RESEARCH EXPERIENCE

3D Garment Generation and Virtual Try-on

- Existing garment-transfer methods often require manual fitting, shared rigs, or predefined correspondences, limiting reuse across diverse characters. Developed LUIVITON to automatically fit rest-posed 3D garments to posed humanoids through SMPL-based correspondence, registration, and neural cloth simulation. [\[SIGGRAPH 2026 / TOG\]](#)
- Sewing patterns vary widely in panel geometry and layout, while discrete representations limit differentiable editing. Trained a latent flow-matching model over implicit-panel codes and implemented guided completion and simulation-based refitting, enabling pattern generation, completion from partial panels, and adaptation across body shapes. [\[SIGGRAPH 2026\]](#)

PUBLICATIONS

01 Learned Universal Interoperable Virtual Try-on

Cong Cao, Xianhang Cheng, Jingyuan Liu, Yujian Zheng, Zhenhui Lin, Ren Li, Meriem Chkir, Hao Li.

ACM Transactions on Graphics 45(4), Article 156, 2026 (SIGGRAPH 2026). [Paper](#)

02 Learning Sewing Patterns via Latent Flow Matching of Implicit Fields

Cong Cao*, Ren Li*, Corentin Dumery, Hao Li.

ACM SIGGRAPH 2026 Conference Papers, Article 179, 2026. [Paper](#)

PUBLICATIONS CONTINUED

03 Spatio-Temporal Garment Reconstruction Using Diffusion Mapping via Pattern Coordinates

Yingxuan You*, Ren Li*, Corentin Dumery, **Cong Cao**, Hao Li, Pascal Fua.

IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI), accepted 2026. [Paper](#)

04 Single View Garment Reconstruction Using Diffusion Mapping Via Pattern Coordinates

Ren Li, **Cong Cao**, Corentin Dumery, Yingxuan You, Hao Li, Pascal Fua.

ACM SIGGRAPH 2025 Conference Papers, 2025. [Paper](#)

* Equal contribution.

INDUSTRY EXPERIENCE

Yalla Group Limited

📅 May 2023 – Jul. 2023

Research and Development Intern

📍 Dubai, UAE

- Fine-tuned CLIP on annotated real-world chatroom images to build an image classifier for content filtering.

TECHNICAL EXPERTISE

Programming and tools: Python, PyTorch, Marvelous Designer, Blender.

Generative modeling: latent flow matching, implicit neural fields, conditional generation, latent-space optimization.

3D vision and graphics: shape correspondence, SMPL/SMPL+D registration, geometry processing, neural cloth simulation, garment reconstruction and virtual try-on.

TEACHING EXPERIENCE

Teaching Assistant, MBZUAI

CV804 3D Geometry Processing

📅 Jan. – May 2026

CV802 Advanced 3D Computer Vision

📅 Aug. – Dec. 2025

HONORS AND AWARDS

Diligence Scholarship, 2020 | Annual University Scholarship, 2019